

Mushroom Image Classification Using Deep Learning

Team 8: Chiu-Chiu (JoJo) Lin, Nicholas Shor, Joanna Tam, Xuewen Yang

Background: Mushroom identification is an important problem in ecology, food safety, and environment studies. Correct identification of mushrooms based on their appearance is challenging since many mushrooms look similar, and some mushrooms that look harmless can be toxic or fatal. This will make manual identification difficult, especially for non-experts who lack the knowledge to distinguish mushroom species accurately. An automated method for identifying mushrooms based on appearance can be useful.

Problem Definition: In this project, we aim to build an image-based model for mushroom edibility classification using ML techniques that assists outdoor enthusiasts in identifying whether a mushroom is safe to eat. The input of our model will be an image of a mushroom in PNG, JPG, or JPEG format. The model will predict the edibility level of the mushroom as edible, conditionally edible, poisonous, or deadly.

Motivation: The core motivation is “Is it safe to eat this mushroom?” This project will classify mushrooms using computer vision to build a tool that empowers those who like camping and nature. The [Mushroom Classification Dataset](#) from Kaggle, with 13.87 GB, is available for these classification models. Each mushroom image is labeled by species and edibility level, enabling robust model training and evaluation.

Literature Review: Previous research on mushroom edibility prediction has mainly used either tabular data or images. The UCI Mushroom Dataset was used to train decision trees, random forests, and other classical models to predict based on physical characteristics such as cap shape, gill color, and odor ([UCI, 1990](#)). More recent studies have applied deep learning to classify edible vs. poisonous mushrooms from images using CNNs, achieving high accuracy on controlled datasets ([Capel et al., 2019](#)). Based on those prior approaches, this project will use a larger labeled image dataset that combines visual features from images with species-level metadata to directly classify and predict 4 classes rather than just 2: edible, conditionally edible, poisonous, or deadly mushrooms.

Approach: We will perform a stratified train-test split on the 104,000 mushroom images, reserving a subset of the training set as a validation set to evaluate the performance of our model. The training data will be used to fine-tune a pre-trained CNN model, such as ResNet or EfficientNet.

Success Criteria: We will use Macro-average F1-Score, precision, and recall to evaluate the performance of our model. The confusion matrix will also be used to understand the specific errors.